

Problem Statement

AI-Assisted Assessment of Medial Meniscus Thickness and Patient-Specific Knee Implant Sizing

Knee osteoarthritis (OA) is one of the most common degenerative joint disorders and is a major cause of pain, stiffness, reduced mobility, and disability among adults. The condition is characterized by progressive changes in the structures of the knee joint, including the articular cartilage, menisci, subchondral bone, and surrounding tissues. Early identification and quantitative assessment of these structural changes can support clinical decision-making and help in planning appropriate treatment. However, the evaluation of knee osteoarthritis from medical images is often dependent on manual assessment by radiologists and orthopedic specialists, making the process time-consuming and potentially subject to inter-observer variation.

The medial meniscus plays an important role in load distribution, shock absorption, joint stability, and protection of the articular cartilage. Structural changes in the medial meniscus, including variations in its thickness, may be associated with the progression and severity of knee osteoarthritis. However, accurately identifying and measuring the medial meniscus between the femur and tibia from medical imaging can be challenging because of variations in patient anatomy, image quality, imaging orientation, and disease-related structural changes. Furthermore, meniscus measurements may differ between male and female populations and across different age groups. There is therefore a need for an automated and quantitative approach that can measure medial meniscus thickness consistently and investigate its relationship with osteoarthritis.

The proposed solution is an **AI-assisted medical imaging analysis platform** capable of automatically identifying the major anatomical structures of the knee, particularly the femur, tibia, and medial meniscus, from suitable knee imaging data. The system will locate the medial meniscus in the joint space between the femur and tibia and measure its thickness at predefined anatomical locations. Instead of relying only on visual assessment, the proposed system will generate quantitative measurements that can be analyzed against clinically labelled osteoarthritis cases.

The measured medial meniscus thickness will be associated with important patient characteristics such as **age, sex, and osteoarthritis status**. The system will enable comparison of meniscus thickness between individuals diagnosed with osteoarthritis and those without osteoarthritis. It will also facilitate separate analysis of male and female populations to determine whether measurable differences exist between the groups. Such quantitative analysis can help identify patterns and potential associations between medial meniscus morphology and knee osteoarthritis.

The system is not intended to replace clinical diagnosis. Instead, it is designed as a **clinical decision-support and research tool** that can assist medical professionals by providing objective and reproducible measurements from medical imaging. By reducing the amount of manual measurement required, the proposed approach can potentially improve efficiency and provide standardized anatomical measurements for further clinical evaluation and research.

Patient-Specific Femoral and Tibial Measurement for Knee Implant Sizing

In addition to osteoarthritis assessment, accurate anatomical measurement is essential for patients undergoing **total knee arthroplasty (TKA)**. TKA involves replacing damaged portions of the knee joint with artificial femoral and tibial components. Selecting an appropriately sized implant is critical because an implant that is too large or too small may result in complications such as component overhang, under-coverage, soft-tissue irritation, improper alignment, altered knee biomechanics, and reduced patient satisfaction.

Currently, implant sizing involves obtaining anatomical measurements from medical imaging and comparing these measurements with available implant specifications. The process may involve manual measurements, surgical planning software, and surgeon experience. Differences in patient anatomy, sex, age, bone morphology, and imaging quality can make standardized implant selection challenging. An automated approach capable of extracting patient-specific femoral and tibial dimensions could support more consistent pre-operative planning.

The proposed platform will therefore incorporate an additional module for **automated femoral and tibial anatomical measurement and implant size estimation**. Using knee imaging data, the AI model will identify and segment the femur and tibia and extract relevant anatomical dimensions required for implant planning. These measurements may include parameters such as femoral width, femoral anteroposterior dimensions, tibial width, tibial anteroposterior dimensions, and other clinically relevant measurements depending on the selected implant system and available imaging modality.

The extracted measurements will then be compared against a structured **knee implant sizing database** containing the dimensional specifications of available femoral and tibial implant components. Based on the patient's measured anatomy, the system can identify the closest matching implant size or provide a ranked set of suitable implant sizes for clinical review.

The objective is to create a unified AI-assisted platform that connects **medical image analysis, anatomical measurement, osteoarthritis assessment, and patient-specific implant planning**. The system will transform medical imaging data into measurable anatomical information rather than relying solely on qualitative visual interpretation.

Proposed Integrated Approach

The proposed system will consist of two major analytical modules. The first module will focus on **medial meniscus analysis and osteoarthritis assessment**. It will detect the femur, tibia, and medial meniscus, calculate meniscus thickness at predefined locations, and compare the measurements with clinically labelled OA and non-OA cases. Statistical and AI-based analysis can then be used to study the relationship between meniscus thickness, osteoarthritis, age, and sex.

The second module will focus on **femoral and tibial anatomical measurement for implant sizing**. It will segment the relevant bone structures, calculate predefined anatomical dimensions, and compare these measurements against an implant database to identify appropriate implant size options.

The combined system can provide a structured output containing anatomical measurements, meniscus thickness, osteoarthritis-related analysis, patient demographic information, and potential implant size recommendations. A visualization interface can display the detected structures and measurement locations directly on the medical image, allowing clinicians and researchers to verify the AI-generated measurements.

Overall, the proposed solution addresses two important challenges in knee healthcare: **quantitative assessment of structural changes associated with osteoarthritis and accurate patient-specific anatomical measurements for knee implant planning**. By applying artificial intelligence and medical image processing, the system aims to reduce manual measurement effort, improve measurement consistency, support comparative analysis across male and female populations, and provide data-driven assistance during pre-operative planning.

The proposed platform can ultimately serve as a foundation for developing an intelligent knee analysis system capable of supporting **early research into osteoarthritis-related structural changes and personalized total knee arthroplasty planning**, while keeping final diagnosis and treatment decisions under the supervision of qualified medical professionals.